National Institute of Technology Karnataka Surathkal
Department of Information Technology

IT 252 DATABASE SYSTEMS

More SQL: Complex Queries, Triggers, Views, and Schema Modification

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Course Outline

Course Plan: Theory:

Part A: Parallel Computer Architectures

Week 1: Introduction to to the course, highlighting the data and databases, and related basic concepts. Advantages and need for a database system.

Week 2: Attributes, tuples, relational schema, conceptual model, introduction to SQL-DML, DDL, creating relations/ tables, access and manipulate data.

Week 3,4: Data model, relational model, and SQL, relational data model and relational database constraints, SQL data definition and data types, specifying constraints, basic retrieval queries, complex quires, triggers, views, schema modification.

Week 5-6: ER model development, entity types/sets, attribute, relationship types/sets, simple employee database conceptual design using ER concepts, ER to relational mapping algorithm, EER model.

Course Outline

Week 7,8 : Functional dependency definition, need of normalization, anomalies and redundant information in tuples, normalization steps (INF, 2NF, 3NF), BCNF, introduction to the higher normal forms.

Week 9: Procedural query languages: relational algebra and SQL-SELECT, PROJECT, RENAME, binary and JOIN operations with examples.

Non-Procedural query languages: relational calculus, and SQL-tuple relational calculus, existential a universal quantifier with examples.

Week 10,11 : Transaction management, schedule and serializability, concurrency control,2-phase lock, recovery mechanism: undo/redo values.

Week 12,13: Disk storage- basic file structures, ordered/unordered. binary search, hashing, indexing, importance of indexes, primary and secondary indexing methods, clustered, B tree, B++ trees.

Week 14: Current trends in database system, introduction to data warehousing, data mining.

Practical: MySQL commands

Project: Team of 2 or 3 members

More Complex SQL Retrieval Queries

- Additional features allow users to specify more complex retrievals from database: such as Nested queries, joined tables, and outer joins (in the FROM clause), aggregate functions, and grouping.

Comparisons Involving NULL and Three-Valued Logic

- Meanings of `NULL`
  - **Unknown value**
  - **Unavailable or withheld value**
  - **Not applicable attribute**
- Each individual `NULL` value considered to be different from every other `NULL` value
- SQL uses a three-valued logic:
  - `TRUE`, `FALSE`, and `UNKNOWN` (like Maybe)
- **NULL = NULL comparison is avoided**

meanings of NULL:

1. **Unknown value.** A person's date of birth is not known, so it is represented by NULL in the database. An example of the other case of unknown would be NULL for a person's home phone because it is not known whether or not the person has a home phone.
2. **Unavailable or withheld value.** A person has a home phone but does not want it to be listed, so it is withheld and represented as NULL in the database.
3. **Not applicable attribute.** An attribute LastCollegeDegree would be NULL for a person who has no college degrees because it does not apply to that person.

Comparisons Involving NULL

- SQL allows queries that check whether an attribute value is NULL. Rather than using = or <> to compare an attribute value to NULL, SQL uses the comparison operators **IS** or **IS NOT**.
- This is because SQL considers each NULL value as being ***distinct*** from every other NULL value, so equality comparison is not appropriate.

Query 18. Retrieve the names of all employees who do not have supervisors.

```
Q18:    SELECT    Fname, Lname
        FROM      EMPLOYEE
        WHERE     Super_ssn IS NULL;
```

Aggregate Functions in SQL

- Used to summarize information from multiple tuples into a single-tuple summary
- Built-in aggregate functions
  - **`COUNT`**, **`SUM`**, **`MAX`**, **`MIN`**, and **`AVG`**
- **Grouping**
  - Create subgroups of tuples before summarizing
- To select entire groups, `HAVING` clause is used
- Aggregate functions can be used in the `SELECT` clause or in a `HAVING` clause

The **COUNT** function returns the number of tuples or values as specified in a query.

Aggregate Functions in SQL

Query 19. Find the sum of the salaries of all employees, the maximum salary, the minimum salary, and the average salary

Renaming Results of Aggregation

- Following query returns a single row of computed values from EMPLOYEE table:

```
Q19:    SELECT  SUM (Salary), MAX (Salary), MIN (Salary), AVG
                (Salary)
        FROM    EMPLOYEE;
```

- The result can be presented with new names:

```
Q19A:   SELECT  SUM (Salary) AS Total_Sal, MAX (Salary) AS
                Highest_Sal, MIN (Salary) AS Lowest_Sal, AVG
                (Salary) AS Average_Sal
        FROM    EMPLOYEE;
```

Aggregate Functions in SQL (cont'd.)

- NULL values are discarded when aggregate functions are applied to a particular column

Query 20. Find the sum of the salaries of all employees of the 'Research' department, as well as the maximum salary, the minimum salary, and the average salary in this department.

```
Q20:  SELECT  SUM (Salary), MAX (Salary), MIN (Salary), AVG (Salary)
      FROM    (EMPLOYEE JOIN DEPARTMENT ON Dno=Dnumber)
      WHERE   Dname='Research';
```

Queries 21 and 22. Retrieve the total number of employees in the company (Q21) and the number of employees in the 'Research' department (Q22).

```
Q21:  SELECT  COUNT (*)
      FROM    EMPLOYEE;

Q22:  SELECT  COUNT (*)
      FROM    EMPLOYEE, DEPARTMENT
      WHERE   DNO=DNUMBER AND DNAME='Research';
```

The **JOIN** operator is used to combine rows from EMPLOYEE table with the DEPARTMENT table.

Aggregate Functions in SQL

Working of Query 20:

- EMPLOYEE Table:

EmpID	Name	Dno	Salary
1	Alice	10	5000
2	Bob	20	7000
3	Charlie	10	6000

- DEPARTMENT Table:

Dnumber	Dname
10	Research
20	Marketing

When JOIN condition is applied (EMPLOYEE.Dno = DEPARTMENT.Dnumber), only employees with Dno = 10 will be included because the WHERE clause further filters these to just employees in the "Research" department.

Hence, the aggregation functions (SUM, MAX, MIN,AVG) would be applied to just Alice and Charlie, whose Dno is 10 (**Research** department), calculated as follows.

SUM(Salary) = 5000 + 6000 = 11000
MAX(Salary) = 6000
MIN(Salary) = 5000
AVG(Salary) = (5000 + 6000) / 2 = 5500

Grouping: The GROUP BY Clause

- **Partition** relation into subsets of tuples
  - Based on **grouping attribute(s)**
  - Apply function to each such group independently
- `GROUP BY` clause
  - Specifies grouping attributes
- COUNT (*) counts the number of rows in the group

Examples of GROUP BY

- The grouping attribute must appear in the SELECT clause:

```
Q24:    SELECT      Dno, COUNT (*), AVG (Salary)
        FROM        EMPLOYEE
        GROUP BY    Dno;
```

- If the grouping attribute has NULL as a possible value, then a separate group is created for the null value (e.g., null Dno in the above query)
- GROUP BY may be applied to the result of a JOIN:

```
Q25:    SELECT      Pnumber, Pname, COUNT (*)
        FROM        PROJECT, WORKS_ON
        WHERE       Pnumber=Pno
        GROUP BY    Pnumber, Pname;
```

- In **Q24**, the EMPLOYEE tuples are partitioned into groups. Each group contains the employees who work in the same department.
- The COUNT and AVG functions are applied to each such group of tuples. Notice that the SELECT clause includes only the grouping attribute and the aggregate functions to be applied on each group of tuples.

Grouping: The GROUP BY and HAVING Clauses (cont'd.)

- **`HAVING`** clause
  - Provides a condition to select or reject an entire group:
- **Query 26.** For each project *on which more than two employees work*, retrieve the project number, the project name, and the number of employees who work on the project.

```
Q26:    SELECT      Pnumber, Pname, COUNT (*)
        FROM        PROJECT, WORKS_ON
        WHERE       Pnumber=Pno
        GROUP BY    Pnumber, Pname
        HAVING      COUNT (*) > 2;
```

Combining the WHERE and the HAVING Clause

- Consider the query: we want to count the *total* number of employees whose salaries exceed $40,000 in each department, but only for departments where more than five employees work.

- INCORRECT QUERY:

```
SELECT     Dno, COUNT (*)
FROM       EMPLOYEE
WHERE      Salary>40000
GROUP BY   Dno
HAVING     COUNT (*) > 5;
```

How? Analyze

Practice Queries:

Query 1:

Consider the relational table given below and assess about the following SQL queries.
Employee (Empno, Name, Department, Salary). (3)

(i) List all the employees whose name starts with the letter 'L'. (2)
(ii) Find the maximum salary given to employees in each department. (2)
(iii) Find the number of employees working in 'accounts' department. (2)
(iv) Find the second maximum salary from the table. (2)
(v) Find the employee who is getting the minimum Salary. (2)

Query 2:

a) Write an SQL query for the following :

a) To create a table of Hospital database with minimum 5 fields
b) To insert two records
c) To add new field
d) To display all records.

Query 3: Employee database. Primary keys are underlined

employee (<u>employee-name</u>, street, city)
works (<u>employee-name</u>, company-name, salary)
company (<u>company-name</u>, city)
manages (<u>employee-name</u>, manager-name)

Give an expression in SQL for each of the following queries.

a. Find the names of all employees who work for First Bank Corporation.
b. Find the names and cities of residence of all employees who work for First Bank Corporation.
c. Find the names, street addresses, and cities of residence of all employees who work for First Bank Corporation and earn more than $10,000.
d. Find all employees in the database who live in the same cities as the com panies for which they work.
e. Find all employees in the database who live in the same cities and on the same streets as do their managers.
f. Find all employees in the database who do not work for First Bank Corpo ration.

Query 4:

Consider the following relation schema

Works(Pname,Cname,salary)

Lives(Pname,Street,City)

located_in (Cname, city)

Manager(Pname,Mgrname)

Write the SQL queries for the following

i) Find the names of all persons who live in the city Bangalore.

ii) Retrieve the names of all person of "Infosys" whose salary is between Rs .50000

iii) Find the names of all persons who lives and work in the same city

iv) List the names of the people who work for "Tech M" along with the cities they live in.

v) Find the average salary of "Infosys" persons

Query 5:

Consider the following relations for a database that keeps track of automobile sales in a car dealership:
CAR(Serial_no, Model, Manufacturer, Price).
OPTION(Serail_no, Option_name, Price).
SALE(Salesperson_id, Serail_no, Date, Sale_price)
SALESPERSON(Salesperson_id, Name, Phone).

OPTION refers to some optional equipment installed on an automobile.

First specify the foreign leys for this schema, stating any assumptions you made. Next, populate the relation with a few sample tuples, and then give an example of an insertion in the SALE and SALESPERSON relation that violates the referential integrity constraints and of another insertion that does not.

Query 6:

Q.4. A company database needs to store information about employees (identified by ssn, with salary and phone as attributes), departments (identified by dna, with dname and budget as attributes), and children of employees (with name and age as attributes). Employees work in departments; each department is managed by an employee; a child must be identified uniquely by name when the parent (who is an employee; assume that only one parent works for the company) is known. We are not interested in information about a child once the parent leaves the company. Draw an ER diagram that captures this information.

Query 7:

Consider the following relations for a company database Application:

Employee (Eno, Name, Sex, DOB, Doj, Designation, Basic_Pay, Dept_No)
Department (DeptNo, Name)
Project (ProjNo, Name, Dept_No)
Works for (Eno,ProjNo,Date, Hours)

The attributes specified for each relation is self-explanatory. However, the business rules are stated as follows. A department can control a project. An employee can work on any number of projects on a day.
However, an employee cannot work more than once on a project he/she worked on that day. The primary key are underlined.
(i) Identify the foreign keys, Develop DDL to implement the above schema.
(ii) Develop an SQL query to list the department number and the number of Employee in each department.
(iii) Develop a View that will keep track of the department number, the number of employees in the department, and the total basic pay expenditure for each department.
(iv) Develop an SQL query to list the details of employees who have worked in more than three projects on a day.

Query 8: Painting Hire Business

System Description:

A local businesswoman has decided to start her own Internet business, called Masterpieces Ltd, hiring paintings to private individuals and commercial companies. Because of your reputation as a database designer, she has called upon your services to design and implement a database to support her new business. At the initial planning meeting, to discuss the design, the **following user requirements** were requested.

The system must be able to manage the details of customers, paintings and those paintings currently on hire to customers. Customers are categorized as B (bronze), S (silver), G (gold) or P (platinum). These categories entitle a customer to a discount of 0%, 5%, 10% or 15% respectively.

Customers often request paintings by a particular artist or theme (eg animal, landscape, seascape, naval, still-life, etc). Over time a customer may hire the same painting more than once. Each painting is allocated a customer monthly rental price defined by the owner. The owner of the painting is then paid 10% of that customer rental price. Any paintings that are not hired within six months are returned to the owner. However, after three months, an owner may resubmit a returned painting. Each painting can only have one artist associated with it. Several reports are required from the system. Three main ones are:

1. For each customer, a report showing an overview of all the paintings they have hired or are currently hiring.
2. For each artist, a report of all paintings submitted for hire .
3. For each artist, a returns report for those paintings not hired over the past six months.

Remember to identify **key attributes** and **any foreign key** attributes.

Query 9:

Consider the following relational database:

Customer(cno, cname, city)
Quotation(qno, qdate, description, amt_quoted, cno)

Customer and Quotation are related with one-to-many relationship. Write MySQL queries for the following:
i. Display customer names having quotation for 'LCD'
iii. List all the customers bearing quotation dated '20-May-2020'
iii. List all the customers who live in 'U.P.' or 'M. P'.
iv. Display customers of amt_quoted as Rs. 15,000.

Query 10:

a. Write SQL Statements the update the following table STUDENT with new records.

STUDENTS

Student – ID	Student – name	Class	DOB
01	M. ARSLAN	MECOM	12-Apr-82
02	JAVARIA AKRAMI	MIT	18-Sep-81
03	FALAK SHER	BIT	15-Mar-83
04	FATIMA	MECOM	10-Oct-82

b. Write a SQL Statement, which will display the all students who born in year 1981, from the table STUDENT assuming that STUDENT contains records about1500 students.

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ER Diagram:1

(*a*) An insurance agent sells insurance policies to clients. Policies can be of different types such as Vehicle Insurance, Life Insurance, Accident Insurance etc. The agent collects monthly premiums on the policies in the form of cheques of local bank. Database should provide the various details to the user.

(*i*) Identify all entities.

(*ii*) Identify all relations.

(*iii*) E-R Diagram.

ER Diagram:2

a. Draw an E-R Model according to following Case study with assumptions. You may make your own assumptions if needed. After identifying Entities, some attributes are given, other you can assume according to your assumptions.

I. A company "Ahsan Corporation" deals in selling products to its customers after purchasing products from different vendors, vendors have attributes like vendor id, vendor name, address, phone number etc.

II. Company sends purchase order to a vendor and provides the list of items to purchase by the company.

III. An item may be purchased from different vendors.

IV. Item are stored in the stock after they ate received from vendor.

V. Customers send Ahsan Corporation, sales order with the list of products to buy from Ahsan Corporation.

VI. Products are sent to Customers with the invoice after receiving sales order from customers. Invoice has attributes of invoice number, data etc.

VII. Customers make the payments according to invoice after receiving the invoice and make payments on some data.

THANK YOU